

Dr Peter C. Allen

Postdoctoral Researcher - Garvan Institute of Medical Research
Conjoint Lecturer - UNSW Medicine and Health
p.allen@garvan.org.au · Sydney, NSW, Australia

PROFESSIONAL SUMMARY

Biomedical researcher with specialised expertise in single-cell genomics, epigenetics, and computational analysis of immune and respiratory disease. PhD-trained in genetics with subsequent postdoctoral experience leading computational analyses of interstitial lung disease (ILD) at one of Australia's leading medical research institutes. Author of the first single-cell bronchoalveolar lavage atlas in ILD and contributor to multiple high-impact manuscripts in *Nature* and other peer-reviewed journals. Recipient of competitive national research funding and recognised with a national society award for excellence in oral presentation.

EDUCATION

2018 – 2023	Doctor of Philosophy in Genetics University of Alabama at Birmingham, USA Dissertation research conducted at HudsonAlpha Institute for Biotechnology
2012 – 2016	Bachelor of Science in Biology Trevecca Nazarene University, Nashville, TN, USA

CURRENT APPOINTMENTS

Sept 2023 – Present	Postdoctoral Researcher Translational Genomics Program, Garvan Institute of Medical Research, Sydney Supervisor: Professor Joseph Powell - Lead computational analyses of single-cell multi-omic data in interstitial lung disease (ILD) and idiopathic pulmonary fibrosis (IPF). - Coordinate multi-institutional collaborative studies, including an international collaboration with Professor Nick Banovich (TGen, USA) providing access to a 200-patient IPF lung explant multiome dataset. - Mentor junior researchers on experimental study design and advanced computational methodologies. - Lead author on the first single-cell bronchoalveolar lavage atlas in ILD (Allen & Zhang et al., <i>PLOS ONE</i>, 2026).
Aug 2024 – Present	Conjoint Lecturer School of Biomedical Sciences, UNSW Medicine and Health, Sydney

PRIOR EMPLOYMENT

2018 – 2023	PhD Candidate Researcher HudsonAlpha Institute for Biotechnology, Huntsville, AL, USA - Researched the epigenetics of autoimmune diseases including Systemic Sclerosis and Systemic Lupus Erythematosus. - Developed methylation-specific assays to identify lupus-specific signals in patient blood samples. - Created bioinformatic pipelines to analyse genome-wide methylation and expression data.
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| 2017 – 2018 | Cell Manufacturing Associate
Cognate Biosciences, Memphis, TN, USA <ul style="list-style-type: none"> Modified immune cell populations to target the Epstein-Barr virus antigen in patients with childhood lymphoma. |
| 2016 – 2017 | Allograft Processing Technician
Derma Sciences Inc., USA <ul style="list-style-type: none"> Processed amniotic and chorionic tissues from postnatal placentas for use as allografts for lesions and eye injuries. |

AWARDS AND HONOURS

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| 2026 | TSANZ Cell, Immunology and Molecular Biology of the Lung Special Interest Group Award — Best Oral Presentation (\$1{,}000 AUD)
Awarded by the Thoracic Society of Australia and New Zealand for the presentation: “Deconvoluting fibrotic niches in idiopathic pulmonary fibrosis lungs using single-cell multiomics,” TSANZSRS Annual Scientific Meeting, Perth. |
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COMPETITIVE RESEARCH FUNDING

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| 2026 | CRE in Pulmonary Fibrosis Kickstarter Grant — \$100,000
Role: Chief Investigator (CI Allen) |
| 2026 | CRE in Pulmonary Fibrosis Travel Grant — \$2,000
Role: Chief Investigator (CI Allen) |

PUBLICATIONS

*Author name shown in **bold**. Co-first authorship indicated by asterisk (*).*

Peer-Reviewed Journal Articles

- Allen P***, Zhang LY*, Zissel G, Lutzky VP, Apte SH, Groves PL, Tan ME, De Silva T, Spenceley ER, McCloy RA, et al. Single-cell transcriptomics of bronchoalveolar lavage reveals divergent macrophage subpopulations and trajectories in interstitial lung disease. *PLOS ONE* (2026). DOI: 10.1371/journal.pone.0347852.
- Idrees S, Chen H, Sadaf T, Rehman SF, Johansen MD, Paudel KR, Liu G, Wang Y, Luecken MD, Hortle E, Philp AS, Budden KF, O'Rourke M, Kaiko GE, Lucas SEM, Dickinson JL, **Allen PC**, Powell JE, Zhang LY, Chambers DC, et al. Multi-omics to study chronic respiratory diseases and viral infections. *European Respiratory Review* 35(179) (2026).
- Tanudisastro H, Cuomo A, Weisburd B, Welland M, Spenceley E, Franklin M, Xue A, Bowen B, Wing K, ... **Allen P**, et al. Tandem repeat variation shapes immune cell type-specific gene expression. *Nature* (in press, 2025).
- Henry A, Senabouth A, ... **Allen PC**, et al. Single-cell genetics identifies cell type-specific causal mechanisms in complex traits and diseases. *Nature* (under review, 2025).
- Xue A, Fan J, ... **Allen PC**, et al. Genetic regulation of cell type-specific chromatin accessibility shapes immune function and disease risk. *Nature* (under review, 2025). Preprint: doi.org/10.1101/2025.08.27.25334533.
- Allen P**, et al. Genome-wide DNA methylation analysis implicates enrichment of interferon pathway in African American patients with Systemic Lupus Erythematosus and European Americans with lupus nephritis. *Journal of Autoimmunity* (2023).

7. **Allen P**, et al. Distinct genome-wide DNA methylation and gene expression signatures in classical monocytes from African American patients with systemic sclerosis. *Clinical Epigenetics* (2023).

INVITED TALKS AND CONFERENCE PRESENTATIONS

Invited Seminars

March 2026 **“Understanding ILD using single-cell genomics”**
Harry Perkins Institute of Medical Research, Perth, Australia
Hosts: Professors Steve Mutsaers and Cecilia Prêle

Oral Presentations

March 2026 **“Deconvoluting fibrotic niches in idiopathic pulmonary fibrosis lungs using single-cell multiomics”**
TSANZSRS Annual Scientific Meeting, Perth, Australia
Awarded TSANZ Cell, Immunology and Molecular Biology of the Lung SIG Award for Best Oral Presentation.

March 2026 **“Beyond the sum of its parts: a new framework for understanding fibrotic lung disease”**
TSANZSRS Annual Scientific Meeting Postgraduate Workshop, Perth, Australia

Posters

2025 Poster Presentation, Oz Single Cell Conference, Sydney, Australia

2019 Poster Presentation, American College of Rheumatology Annual Meeting, USA. Topic: Epigenetic markers identified in Systemic Sclerosis.

Other Speaking Engagements

2024 **Speaker**, Garvan Day of Immunology, Garvan Institute of Medical Research, Sydney. Delivered key lecture on the importance of immunology research; panellist for high-school careers Q&A session.

2024 **Session Chair (Immunology)**, St Vincent’s Sydney Health Innovation Precinct, Sydney.

2021 – 2023 **Guest Lecturer**, Oakwood University, Huntsville, AL, USA. Lectures on genetics and bioinformatics for senior university students.

PROFESSIONAL SERVICE

Peer Review

- Reviewer, *Clinical Epigenetics*.

Workshop and Event Organisation

2025 **Host**, Single Cell Oz Hackathon, Sydney.

2024 **Workshop Host and Presenter**, CREATE Genomics Workshop. Led workshops on genomic techniques and data analysis.

RESEARCH COLLABORATIONS

International

- **Professor Nick Banovich**, TGen, Phoenix, Arizona, USA — collaboration providing access to a 200-patient IPF lung explant multiome dataset.
- **University of North Carolina at Chapel Hill**, USA — Systemic Sclerosis research collaboration (initiated during PhD).

Domestic

- **Queensland Lung Transplant Service Lab**, Brisbane — interstitial lung disease research.
- **Royal Prince Alfred Hospital**, Sydney — interstitial lung disease research.

TECHNICAL EXPERTISE

Computational and bioinformatic: single-cell RNA sequencing analysis; single-cell ATAC sequencing analysis; multi-omic data integration; gene regulatory network inference; differential expression and trajectory analysis; whole-genome and epigenome-wide association studies; pipeline development (Nextflow, Snakemake); high-performance and cloud computing.

Programming: R, Python, Bash, Git.

Wet-lab: methylation-specific assay development; cell-based immune modification (CAR-T background); tissue processing for clinical-grade allograft applications.

Disease areas: interstitial lung disease, idiopathic pulmonary fibrosis, systemic lupus erythematosus, systemic sclerosis, Epstein-Barr virus-related lymphomas.

PROFESSIONAL MEMBERSHIPS

- Lung Foundation of Australia

REFEREES

Available on request.